6.1: Biologists use microscopes and biochemistry to study cells

How can cell biologists investigate the inner workings of a cell, usually too small to be seen by the unaided eye? Before we tour the cell, it will be helpful to learn how cells are studied.

Microscopy

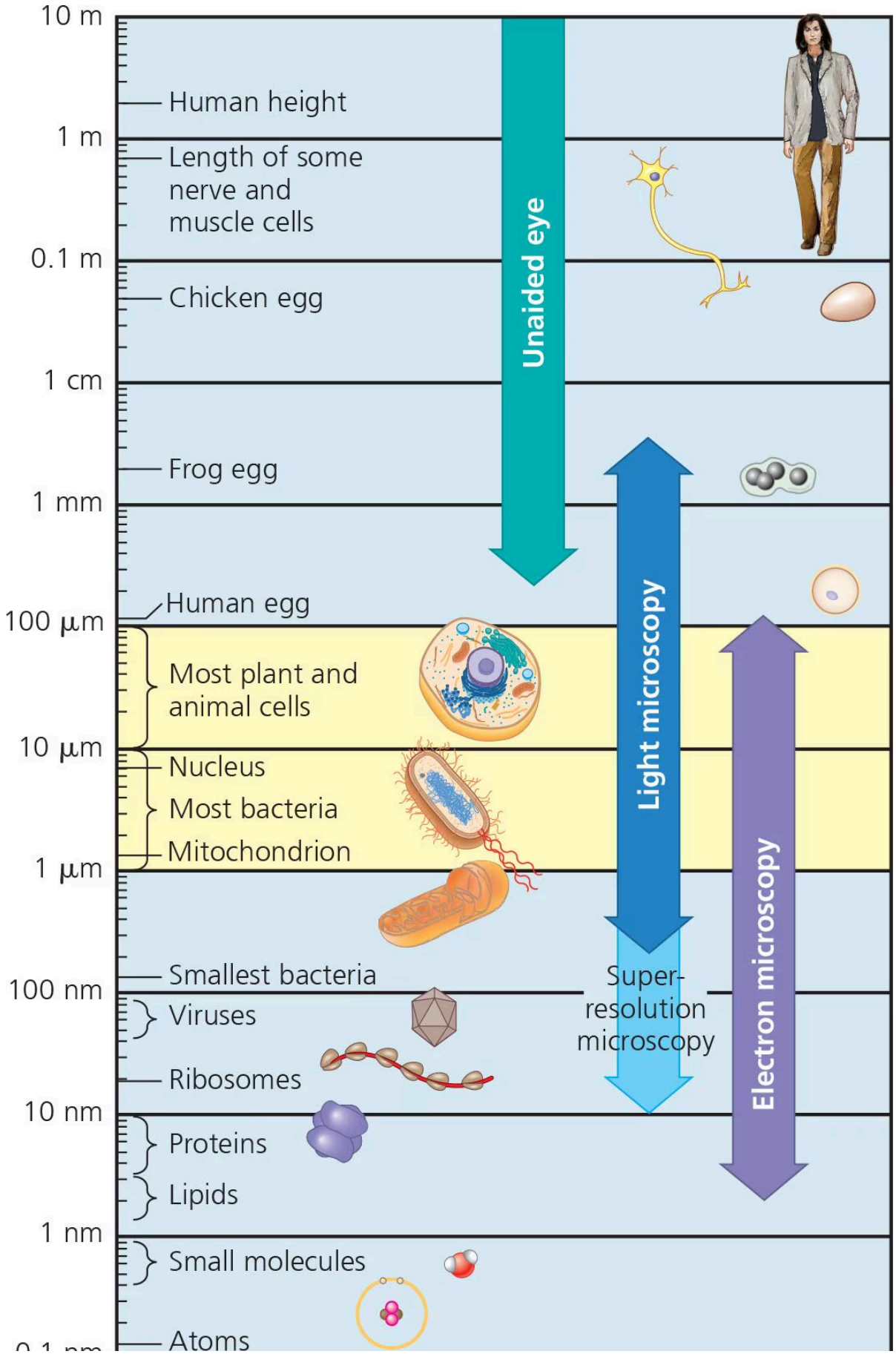
The development of instruments that extend the human senses allowed the discovery and early study of cells. Microscopes were invented in 1590 and further refined during the 1600s. Cell walls were first seen on dead cells of oak bark by Robert Hooke in 1665 and living cells by Antoni van Leeuwenhoek a few years later.

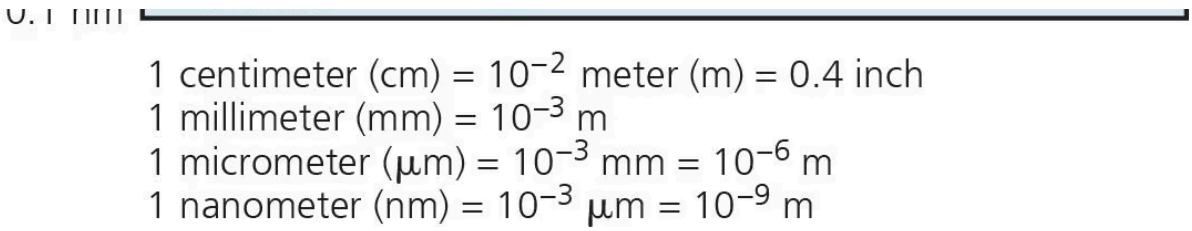
The microscopes first used by Renaissance scientists, as well as the microscopes you are likely to use in the laboratory, are all light microscopes. In a **light microscope(LM)**, visible light is passed through the specimen and then through glass lenses. The lenses refract (bend) the light in such a way that the image of the specimen is magnified as it is projected into the eye or into a camera (see Appendix C).

Three important parameters in microscopy are magnification, resolution, and contrast. *Magnification* is the ratio of an object's image size to its real size. Light microscopes can magnify effectively to about 1,000 times the actual size of the specimen; at greater magnifications, additional details cannot be seen clearly. *Resolution* is a measure of the clarity of the image; it is the minimum distance two points can be separated and still be distinguished as separate points. For example, what appears to the unaided eye as one star in the sky may be resolved as twin stars with a telescope, which has a higher resolving ability than the eye. Similarly, using standard techniques, the light microscope cannot resolve detail finer than about 0.2 micrometer (μm), or 200 nanometers (nm), regardless of the magnification **(Figure 6.2)**. The third parameter, *contrast*, is the difference in brightness between the light and dark areas of an image. Methods for enhancing contrast include staining or labeling cell components to stand out visually. **Figure 6.3** shows some different types of microscopy; study this figure as you read this section.

Figure 6.2

The size range of cells.


10 m
Human height
1 m
Length of some nerve and muscle cells
0.1 m
Chicken egg
1 cm
Frog egg
1 mm
Human egg
100 μm
Most plant and animal cells
10 μm
Nucleus
Most bacteria
Mitochondrion
1 μm
Smallest bacteria
100 nm
Viruses
Ribosomes
10 nm
Proteins
Lipids
1 nm
Small molecules
Atoms
Unaided eye
Light microscopy
Super-resolution microscopy
Electron microscopy



The chart reads as follows. Scale key:

- 1 centimeter = 10 to the minus 2 of a meter = 0.4 inches
- 1 millimeter = 10 to the minus 3 of a meter
- 1 micrometer = 10 to the minus 3 millimeter or 10 to the minus 6 meter
- 1 nanometer = 10 to the minus 3 of a micrometer or 10 to the minus 9 of a meter.

Visible by the unaided human eye:Human height indicated at about 6 meters; length of some muscle and nerve cells (about 0.8 meters); Chicken egg (about 5 centimeters); Frog egg (about 2 millimeters). On the scale of visibility with both the light microscope and the electron microscope are the human egg cell (about 100 micrometers); most plant and animal cells (range from 10 to 100 micrometers); the cell nucleus (about 8 micrometers), cell mitochondria (about 2 micrometers) and most bacteria (range from 1 to 10 micrometers). The smallest bacteria (just over 100 nanometers), viruses (70 to 100 nanometers) and ribosomes (12 nanometers) can be seen with super-resolution light microscopes or an electron microscope. Visible with an electron microscope only are large molecules like proteins (5 to 10 nanometers); and lipids (2 to 5 nanometers). Too small to be seen even with electron microscopes are small molecules (0.5 to 1 nanometer) and atoms (0.2 nanometers).

Most cells are between 1 and 100 μm in diameter (yellow region of chart), and their components are even smaller (see Figure 6.32), as are viruses. Notice that the scale along the left side is logarithmic, to accommodate the range of sizes shown. Starting at the top of the scale with 10 m and going down, each reference measurement marks a tenfold decrease in diameter or length. For a complete table of the metric system, see the back of the book.

Animation

Metric System Review

Here are some prefixes for measurements that are smaller than the base unit:

Prefix	Symbol	Exponent	Multiplier (factor)
centi-	c	10^{-2}	1/100 (one hundredth)
milli-	m	10^{-3}	1/1000 (one thousandth)
micro-	μ	10^{-6}	1/1,000,000 (one millionth)
nano-	n	10^{-9}	1/1,0000,0000,000 (one billionth)
pico-	p	10^{-12}	1/1,000,000,000,000 (one trillionth)

Here are some prefixes for measurements that are larger than the base unit:

Prefix	Symbol	Exponent	Multiplier (factor)
kilo-	k	10^{3}	1000x (one thousand times)
mega-	M	10^{6}	1,000,000x (one million times)
giga-	G	10^{9}	1,000,000,000x (one billion times)
tera-	T	10^{12}	1,0000,0000,000,000x (one trillion times)
peta-	P	10^{15}	1,000,000,000,000,000x (one quadrillion

Play

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giga-	G	10^{9}	1,000,000,000x (one billion times)
tera-	T	10^{12}	1,0000,0000,000,000x (one trillion times)
peta-	P	10^{15}	1,000,000,000,000,000x (one quadrillion

Play

Figure 6.3

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Exploring Microscopy
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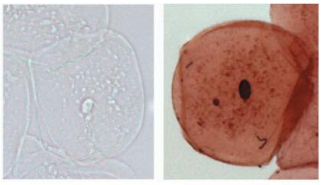
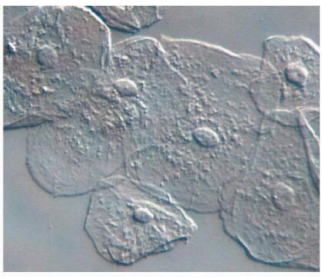
Light Microscopy (LM)

Brightfield. Light passes directly through the specimen. Unstained (left), the image has little contrast. Staining with dyes (right) enhances contrast. Most stains require cells to be preserved, which kills them.



Phase-contrast. Variations in density within the specimen are amplified to enhance contrast in unstained cells; this is especially useful for examining living, unstained cells.

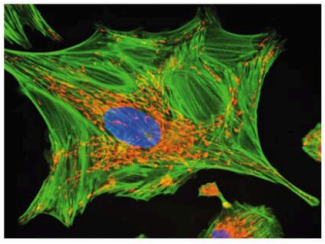
Differential interference contrast (Nomarski). As in phase-contrast microscopy, optical modifications are used to exaggerate differences in density; the image appears almost 3-D.



Human cheek cells

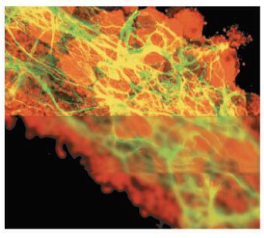
50 µm

Fluorescence. Locations of specific molecules are revealed by labeling the molecules with fluorescent dyes or antibodies, which absorb ultraviolet radiation and emit visible light. In this fluorescently labeled uterine cell, DNA is blue, organelles called mitochondria are orange, and part of the cell's "skeleton" (called the *cytoskeleton*) is green.



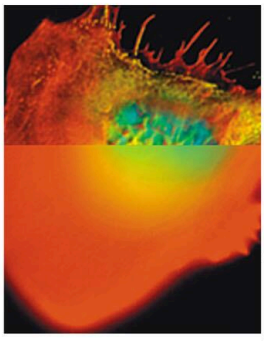
10 µm

Confocal. This image shows two types of fluorescence micrographs: confocal (top) and standard (bottom). (Nerve cells are green, support cells orange, areas of overlap yellow.) In confocal microscopy, a laser is used to create a single plane of fluorescence; out-of-focus light from other planes is eliminated. By capturing sharp images at many different planes, a 3-D reconstruction can be created. A standard fluorescence micrograph is blurry because out-of-focus light is not excluded.



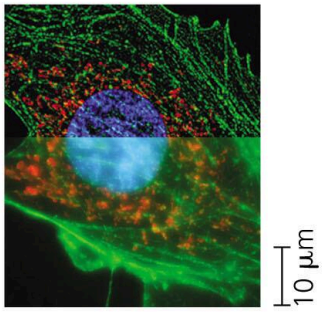
50 µm

Deconvolution. The top of this image of a white blood cell was reconstructed from many blurry fluorescence images at different planes, each processed using deconvolution software. This process digitally removes out-of-focus light and reassigns it to its source, creating a much sharper 3-D image. The bottom is a compilation of standard fluorescent micrographs through the same cell.



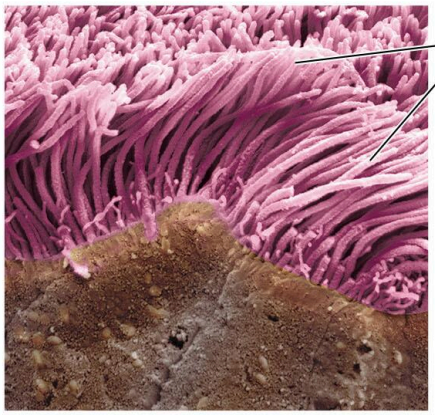
10 µm

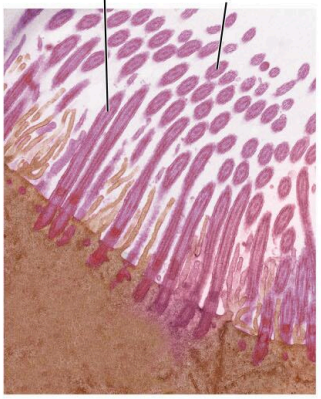
Super-resolution. To make this super-resolution image of a cow aorta cell (top), individual fluorescent molecules were excited by UV light and their position recorded. (DNA is blue, mitochondria red, and part of the cytoskeleton green.) Combining information from many molecules in different places "breaks" the resolution limit, resulting in the sharp image on top. The size of each dot is well below the 200-nm resolution of a standard light microscope, as seen in the confocal image (bottom) of the same cell.



Electron Microscopy (EM)

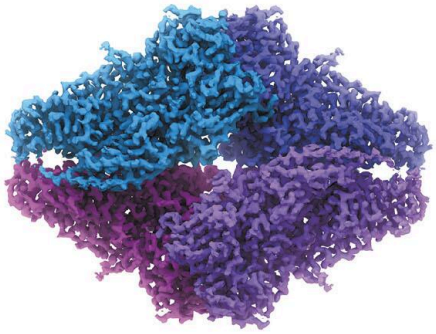
Scanning electron microscopy (SEM). Micrographs taken with a scanning electron microscope show a 3-D image of the surface of a specimen. This SEM shows the surface of a cell from a trachea (windpipe) covered with cell projections called cilia. Electron micrographs are black and white but are often artificially colorized to highlight particular structures, as has been done with all three electron micrographs shown here.





Transmission electron microscopy (TEM). A transmission electron microscope profiles a thin section of a specimen. This TEM shows a section through a tracheal cell, revealing its internal structure. In preparing the specimen, some cilia were cut along their lengths, creating longitudinal sections, while other cilia were cut straight across, creating cross sections.

Cryo-electron microscopy (cryo-EM). Specimens of tissue or aqueous solutions of proteins are frozen rapidly at temperatures less than –160°C, locking the molecules into a rigid state. A beam of electrons is passed through the sample to visualize the molecules by electron microscopy, and software is used to merge a series of such micrographs, creating a 3-D image like the one shown here.



Computer-generated image of the bacterial enzyme β-galactosidase, which breaks down lactose. This image was compiled from more than 90,000 cryo-EM images.

Abbreviations used in figure legends in this text:
LM = Light Micrograph
SEM = Scanning Electron Micrograph
TEM = Transmission Electron Micrograph

Types of microscopy are as follows.

- Brightfield (unstained specimen). Light passes directly through the specimen. Unless the cell is naturally pigmented or artificially stained, the image has little contrast. (The first four light micrographs show human cheek epithelial cells; the scale bar pertains to all four micrographs.)

- Brightfield (stained specimen). Staining with various dyes enhances contrast. Most staining procedures require that cells be fixed (preserved), thereby killing them.
- Phase-contrast. Variations in density within the specimen are amplified to enhance contrast in unstained cells; this is especially useful for examining living, unpigmented cells.
- Differential interference contrast (Nomarski). As in phase-contrast microscopy, optical modifications are used to exaggerate differences in density; the image appears almost 3-D.
- Fluorescence. The locations of specific molecules in the cell can be revealed by labeling the molecules with fluorescent dyes or antibodies; some cells have molecules that fluoresce on their own. Fluorescent substances absorb ultraviolet radiation and emit visible light. In this fluorescently labeled uterine cell, nuclear material is blue, organelles called mitochondria are orange, and the cell's skeleton is green.
- Confocal. The top image is a standard fluorescence micrograph of fluorescently labeled nervous tissue (nerve cells are green, support cells are orange, and regions of overlap are yellow); below it is a confocal image of the same tissue. Using a laser, this optical sectioning technique eliminates out-of-focus light from a thick sample, creating a single plane of fluorescence in the image. By capturing sharp images at many different planes, a 3-D reconstruction can be created. The standard image is blurry because out-of-focus light is not excluded.
- Deconvolution. The top of this split image is a compilation of standard fluorescence micrographs through the depth of a white blood cell. Below is an image of the same cell reconstructed from many blurry images at different planes, each of which was processed using deconvolution software. This process digitally removes out-of-focus light and reassigns it to its source, creating a much sharper 3-D image.
- Super-resolution. On the top is a confocal image of part of a nerve cell, using a fluorescent label that binds to a molecule clustered in small sacs in the cell (vesicles) that are 40 nm in diameter. The greenish-yellow spots are blurry because 40 nm is below the 200-nm limit of resolution for standard light microscopy. Below is an image of the same part of the cell, seen using a new super-resolution technique. Sophisticated equipment is used to light up individual fluorescent molecules and record their position. Combining information from many molecules in different places breaks the limit of resolution, resulting in the sharp greenish-yellow dots seen here. (Each dot is a_40-nm vesicle.)
- Scanning electron microscopy (SEM). Micrographs taken with a scanning electron microscope show a 3-D image of the surface of a specimen. This SEM shows the surface of a cell from a trachea (windpipe) covered with cell projections called cilia. Electron micrographs are black and white but are often artificially colorized to highlight particular structures, as has been done with both electron micrographs shown here.

- Transmission electron microscopy (TEM). A transmission electron microscope profiles a thin section of a specimen. This TEM shows a section through a tracheal cell, revealing its internal structure. In preparing the specimen, some cilia were cut along their lengths, creating longitudinal sections, while other cilia were cut straight across, creating cross sections. Abbreviations used in figure legends in this text: LM = Light Micrograph. SEM = Scanning Electron Micrograph. TEM = Transmission Electron Micrograph

VISUAL SKILLS *When the tissue was sliced, what was the orientation of the cilia in the lower portion of the TEM? The upper portion? Explain how the orientation of the cilia determined the type of sections we see.*

*For suggested answer, see*Appendix A*.*

Until recently, the resolution barrier prevented cell biologists from using standard light microscopy when studying **organelles**, the membrane-enclosed structures within eukaryotic cells. To see these structures in any detail required the development of a new instrument. In the 1950s, the electron microscope was introduced to biology. Rather than focusing light, the **electron microscope(EM)** focuses a beam of electrons through the specimen or onto its surface (see Appendix C). Resolution is inversely related to the wavelength of the light (or electrons) a microscope uses for imaging, and electron beams have much shorter wavelengths than visible light. Modern electron microscopes can theoretically achieve a resolution of about 0.002 nm, though in practice they usually cannot resolve structures smaller than about 2 nm across. Still, this is a 100-fold improvement over the standard light microscope.

The **scanning electron microscope(SEM)** is especially useful for detailed study of the topography of a specimen (see Figure 6.3). The electron beam scans the surface of the sample, usually coated with a thin film of gold. The beam excites electrons on the surface, and these secondary electrons are detected by a device that translates the pattern of electrons into an electronic signal sent to a video screen. The result is an image of the specimen's surface that appears three-dimensional.

The **transmission electron microscope(TEM)** is used to study the internal structure of cells (see Figure 6.3). The TEM aims an electron beam through a very thin section of the specimen, much as a light microscope aims light through a sample on a slide. For the TEM, the specimen has been stained with atoms of heavy metals, which attach to certain cellular structures, thus enhancing the electron density of some parts of the cell more than others. The electrons passing through the specimen are scattered more in the denser regions, so fewer are transmitted. The image displays the pattern of transmitted electrons. Instead of using glass lenses, both the SEM and TEM use electromagnets as lenses to bend the paths of the electrons, ultimately focusing the image onto a monitor for viewing.

Electron microscopes have revealed many subcellular structures that were impossible to resolve with the light microscope. But the light microscope offers advantages, especially in studying living cells. A disadvantage of electron microscopy is that the methods customarily used to prepare the specimen kill the cells and can introduce artifacts, structural features seen in micrographs that do not exist in the living cell.

In the past several decades, light microscopy has been revitalized by major technical advances (see Figure 6.3). Labeling individual cellular molecules or structures with fluorescent markers has made it possible to see such structures with increasing detail. In addition, both confocal and deconvolution microscopy have produced sharper images of three-dimensional tissues and cells. Finally, a group of new techniques and labeling molecules developed in recent years, called *super-resolution microscopy*, has allowed researchers to "break" the resolution barrier and distinguish subcellular structures as small as 10-20 nm across.

A recently developed new type of TEM called cryo-electron microscopy (cryo-EM) (see Figure 6.3) allows specimens to be preserved at extremely low temperatures. This avoids the use of preservatives, allowing visualization of structures in their cellular environment. This method is increasingly used to complement X-ray crystallography in revealing protein complexes and subcellular structures like ribosomes, described later. Cryo-EM has even been used to resolve some individual proteins. The Nobel Prize for Chemistry was awarded in 2017 to the developers of this valuable technique.

Microscopes are the most important tools of *cytology*, the study of cell structure. Understanding the function of each structure, however, required the integration of cytology and *biochemistry*, the study of the chemical processes (metabolism) of cells.

Cell Fractionation

A useful technique for studying cell structure and function is **cell fractionation(Figure 6.4)**, which takes cells apart and separates major organelles and other subcellular structures from one another. The piece of equipment that is used for this task is the centrifuge, which spins test tubes holding mixtures of disrupted cells at a series of increasing speeds, a process called *differential centrifugation*. At each speed, the resulting force causes a subset of the cell components to settle to the bottom of the tube, forming a pellet. At lower speeds, the pellet consists of larger components, and higher speeds result in a pellet with smaller components.

Figure 6.4

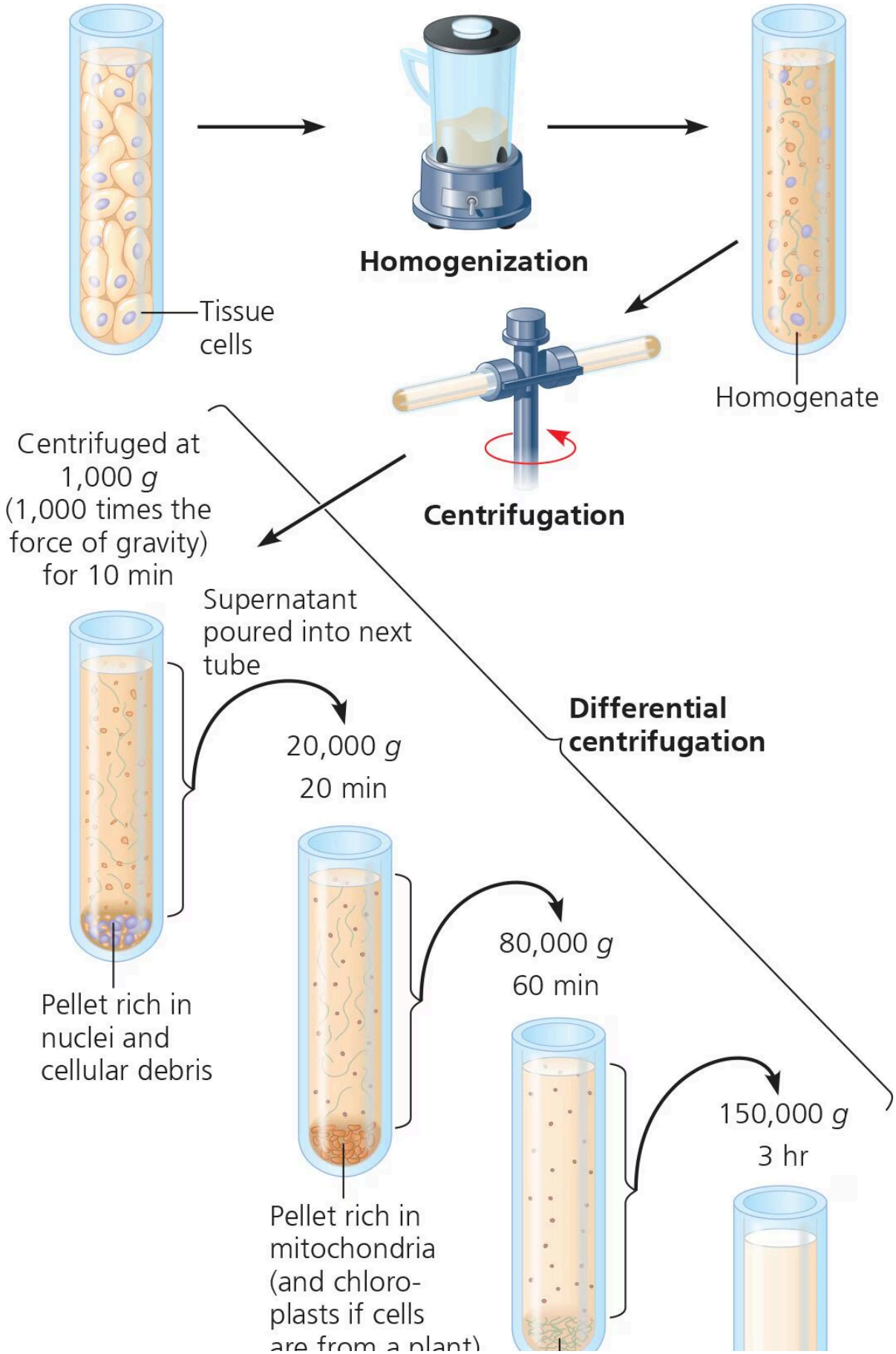
Research Method: Cell Fractionation

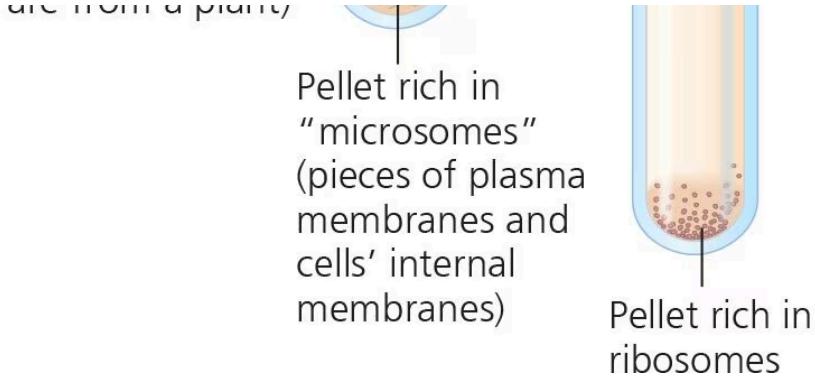
Application

Cell fractionation is used to separate (fractionate) cell components based on size and density.

Technique

Cells are homogenized in a blender to break them up. The resulting mixture (*homogenate*) is centrifuged. The liquid above the pellet (*supernatant*) is poured into another tube and centrifuged at a higher speed for a longer period. This process is repeated several times. This process, called *differential centrifugation*, results in a series of pellets, each containing different cell components.


Tissue cells
Homogenization
Homogenate
Centrifuged at 1,000 g (1,000 times the force of gravity) for 10 min
Centrifugation
Supernatant poured into next tube
Differential centrifugation
20,000 g
20 min
Pellet rich in nuclei and cellular debris
80,000 g
60 min
Pellet rich in mitochondria (and chloro-plasts if cells are from a plant)
150,000 g
3 hr



Tissue cells are homogenized in a blender to create homogenate, which is placed in a centrifuge. It is centrifuged at 1,000 times the force of gravity for 10 minutes, resulting in a pellet in the tube rich in nuclei and cellular debris. The supernatant from the first tube is poured into the next tube, which is centrifuged at 2,000 times the force of gravity for 20 minutes, resulting in a pellet rich in mitochondria and chloroplasts if cells are from a plant. The second tube is poured into a third which is centrifuged at 80,000 times the force of gravity for 60 minutes, resulting in a pellet rich in microsomes, pieces of plasma membranes and cells internal membranes. The third tube is poured into a fourth, which is centrifuged at 150,000 times the force of gravity for 3 hours, resulting in a pellet rich in ribosomes.

Results

In early experiments, researchers used microscopy to identify the organelles in each pellet and biochemical methods to determine their metabolic functions. These identifications established a baseline for this method, enabling today's researchers to know which cell fraction they should collect in order to isolate and study particular organelles.

MAKE CONNECTIONS*If you wanted to study the process of translation of proteins from mRNA, which part of which fraction would you use? (See*Figure 5.22*.)**

*For suggested answer, see*Appendix A*.**

Cell fractionation enables researchers to prepare specific cell components in bulk and identify their functions, a task not usually possible with intact cells. For example, on one of the cell fractions, biochemical tests showed the presence of enzymes involved in cellular respiration, while electron microscopy revealed large numbers of the organelles called mitochondria. Together, these data helped biologists determine that mitochondria are the sites of cellular respiration. Biochemistry and cytology thus complement each other in correlating cell function with structure.